

Model / method	DTI	Proteomics	Cell perturbation		
	BindingDB AUROC ↑	PTPC AP ↑	VCC Top-1 ↑	Norman Top-1 ↑	Tahoe Top-1 ↑
<i>Fixed task models, one laboratory</i>					
TAPB	81.76	–	–	–	–
ProteinTalks-derived head	–	<u>23.97</u>	–	–	–
scDEBART response head	–	–	<u>29.94</u>	25.56	<u>19.40</u>
Qwen direct (1 lab)	<u>88.82</u>	20.21	21.73	14.15	13.43
AI-Scientist-v2 (10 labs)	N/A	37.12	31.63	<u>22.74</u>	23.88
AI-Researcher (10 labs)	N/A	N/A	N/A	N/A	N/A
BioCoLoop (10 labs)	94.60	37.12	31.63	<u>22.74</u>	23.88

Table 1: Matched seed-42 pilot preview. All methods use the same training/search seed 42 and the named task models. Native AI-Scientist-v2 and AI-Researcher controllers are task-adapted with local Qwen, ten laboratories, a twelve-design menu, six proposal slots and 100-round candidates. Scores are multiplied by 100; missing seed-42 results are N/A. Bold and underline mark the best and second-best displayed scores across all five methods, including ties.

Model / method	DTI	Proteomics	Cell perturbation		
	BindingDB AUROC $\uparrow$	PTPC AP $\uparrow$	VCC Top-1 $\uparrow$	Norman Top-1 $\uparrow$	Tahoe Top-1 $\uparrow$
<i>Fixed task models, one laboratory</i>					
TAPB	82.83 ± 1.22	–	–	–	–
ProteinTalks-derived head	–	25.05 ± 4.56	–	–	–
scDEBART response head	–	–	29.94 ± 0.00	25.56 ± 0.00	19.40 ± 0.00
Qwen direct (1 lab)	89.89 ± 1.02	20.13 ± 0.07	21.09 ± 4.38	14.15 ± 0.00	15.42 ± 6.21
AI-Scientist-v2 (10 labs)	N/A	N/A	N/A	N/A	N/A
AI-Researcher (10 labs)	N/A	N/A	N/A	N/A	N/A
BioCoLoop (10 labs)	93.76 ± 1.11	36.99 ± 0.22	31.54 ± 0.08	22.25 ± 0.45	22.39 ± 2.59

Table 2: Preview pending manuscript integration. All methods use the named task models. Native AI-Scientist-v2 and AI-Researcher controllers are task-adapted with local Qwen, ten laboratories, a twelve-design menu, six proposal slots and 100-round candidates. Values are mean $\pm$ sample SD over complete seeds 42–44, multiplied by 100; incomplete cells are N/A. Bold and underline mark the best and second-best displayed means across all five methods, including ties.